Internal Assessment - 01

The Devid Case

Case Background: Mr. Devid's 200MB USB drive was seized in a fraud investigation. The forensic image **Devid_evidence.img** is provided for analysis. The image contains an MBR partition table and a single FAT32 partition. You must examine the image, recover deleted evidence, and extract hidden data.

Q1. Create Copy and Verify Hash *(4 Marks)*

a) Write the command to create a working copy named Devid_working.img from the original image file. (*2 Marks*)
b) Write commands to generate both MD5 and SHA-256 hash values of the original image file. (*1 Marks*)
c) Explain why hash verification is critical before beginning any forensic analysis. (*1 Marks*)

Q2. Analyse Partition and Recover Deleted Files *(4 Marks)*

a) Analyse the partition structure of the provided image. What command would you use to view the partition layout? Based on your analysis, what is the starting sector offset of the main partition? (*1 Marks*)
b) From your analysis, list the **names** of all deleted files you were able to identify also give the command for show all the deleted files. (*1 Marks*)
c) Choose any one deleted file you recovered. Write the command to extract and display its contents. Also, provide a brief summary of what information was found inside that file. *(2 Marks*)

Q3. Image Metadata Analysis *(2 Marks)*

Inside the disk image, you will find a photograph file. Examine this image file to extract its metadata.

a) What is the **physical location (GPS coordinates) device make and model** where this photograph was taken? (*1 Marks*)
b) Why is metadata analysis important in digital forensics investigations? Provide one example of how this information could be used as evidence. (*1 Marks*)